

Large Language Model System Design

Transformers are the dominant architecture behind modern language models. They use self-attention mechanisms to process tokens in parallel.

Accuracy vs Latency Tradeoff

Large models often achieve higher accuracy but require more compute, leading to higher latency.

For real-time applications such as chat systems or retrieval augmented generation pipelines, latency is critical.

Model Comparison

Models such as Mistral 7B are designed to provide strong performance while maintaining lower latency.

Deployment Considerations

When deploying LLM systems, engineers must balance:

- model accuracy
- inference latency
- infrastructure cost
- memory requirements

RAG Systems

Retrieval Augmented Generation systems retrieve relevant documents from external knowledge bases and provide them as context to language models.

This allows systems to answer questions using up-to-date knowledge.